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# Motion Intention Recognition Based on Deep Learning Approaches for EEG Signal Analysis Considering Laban/Bartenieff Movement System

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***Abstract*—Brain-Computer Interface (BCI) based on Motor Imagery (MI) is a key technology for achieving fine-grained motor intention recognition. However, existing studies lack a systematic movement classification paradigm, resulting in insufficient decoding accuracy for complex intentions. To address this issue, this paper introduces the 'Effort' framework of the Laban/Bartenieff Movement System (LBMS), focusing on the 'Light-Strong' classification, and categorizes catching and releasing actions into four types: Light Catching (LC), Light Releasing (LR), Strong Catching (SC), and Strong Releasing (SR). EEG signals were collected from subjects via a non-invasive device through controlled experiments, with light and strong action data groups acquired. After preprocessing (including denoising, band-pass filtering, and data augmentation), Deep Learning (DL) methods were adopted for classification: Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) networks were constructed to verify the discriminability of 'Light-Strong' related EEG features; a hybrid CNN-BiLSTM-Attention model was proposed to realize multi-class action recognition, presenting a novel feature description paradigm for motor intention recognition.**

***Keywords- BCI; Motor Imagery; EEG; Deep Learning; Hybrid Neural Network***

## I. Introduction

Brain-Computer Interface (BCI), a core interdisciplinary field of brain science and engineering technology, enables ‘brain-machine’ collaborative interaction by establishing direct information channels between the brain and external devices[1]. Among mainstream BCI paradigms, Motor Imagery (MI) is an important means of acquiring movement-related neural signals. Subjects can induce characteristic electroencephalographic (EEG) activities in the motor cortex by mentally simulating specific movements (e.g., hand grasping, limb flexion, and extension), providing a signal source for motor intention recognition[2]. However, achieving high classification accuracy remains a significant challenge in practical MI-BCI applications.

A central challenge of achieving accurate neural decoding in BCI research stems precisely from the prevailing use of simple movement classification paradigms, which lacking systematic movement pattern classification and thus cannot meet the demand for fine-grained intention recognition in complex scenarios[3]. Studies have shown that force intensity (weight) and trajectory fluency are correlated with EEG activities in the motor cortex, which can provide a clear direction for feature extraction of EEG signals[4]. To address the aforementioned challenges in motor intention recognition, this paper introduces the LBMS as an auxiliary analytical framework. Proposed systematically by Hungarian movement theorist Rudolf Laban in the early 20th century, LBMS was initially used for the standardized description and teaching of dance movements[4]. This theory constructs a standardized qualitative movement description system from four components, one of which is ‘Effort’, concerning energy intensity and rhythm[5]. A key factor in Laban/Bartenieff Movement System Theory within the Effort component is Weight. Its opposing tendencies (Strong and Light) directly reflect the muscle tension of the movement performer and the interaction mode between the body and gravity, which are highly correlated with the intrinsic motivation behind the movement[6]. Therefore, the introduction of the Effort component provides a comprehensive analytical system that enables a more refined understanding of human movements and further improves the classification accuracy of brain-computer interfaces.

EEG signals require the mining of complex features. Based on signal characteristics and decoding needs, deep learning models (such as CNN, LSTM, and Transformer) have become the current research trend in EEG signal recognition[8][9][10]. To overcome the limitations of single-dimensional feature mining, hybrid architectures such as CNN-LSTM and CNN-Transformer have been proposed. These hybrid architectures simultaneously integrate spatial features and temporal features (or long-term dependency features), adapting to the inherent spatiotemporal coupling characteristics of motor imagery signals and meeting decoding needs in complex scenarios[10][12].

Moreover, in addition to temporal information, the spatial and frequency information of EEG signals also play a critical role in motor intention decoding. Spatial features can accurately characterize the topological distribution differences of activated regions in the motor cortex—for example, movement imagery with different weight tendencies can induce changes in activation intensity in different brain regions of the motor cortex, and the effectiveness of this feature can be verified through extraction methods based on multi-scale spatial segmentation strategies[13]. Frequency features (such as rhythmic changes of α waves [8~12Hz] and β waves [12~30Hz]) can reflect the dynamic rhythmic characteristics of neural activities during movement imagery, and their correlation with movement attributes can be supported by individual-specific frequency band analysis[13][15]. The synergistic integration of the two can effectively compensate for the limitations of a single feature dimension and improve the recognition accuracy and robustness of decoding models[16].

This study adopts a non-invasive Brain-Computer Interface (BCI) combined with time-frequency-spatial domain processing. Two deep learning models, CNN and LSTM, are used to classify the collected data, verifying that the introduction of LBMS improves the model training performance. Meanwhile, a hybrid CNN-BiLSTM-Attention model is constructed, achieving high classification accuracy for the four-class task.

## II. Processing and Classification of EEG Data

### A. EEG data acquisition

In this study, based on the Light-Strong classification within the ‘Effort’ component of the LBMS, catching and releasing actions were categorized into four types: ‘Light Catching (LC)’, ‘Light Releasing (LR)’, ‘Strong Catching (SC)’, and ‘Strong Releasing (SR)’ to investigate brain activity responses under different task conditions. The experiment was conducted with one healthy subject. During the session, the subject first clarified the action types through demonstrations by the experimenters. Upon hearing an acoustic trigger signal, the subject mentally imagined the corresponding action, after which the EEG equipment collected data from 0.8s to 2.0s post-stimulus at a sampling rate of 500Hz. Thus, a data sequence of length 600 (500Hz × 1.2s) was acquired for each trial. The experimental scenario is illustrated in Figure 1.

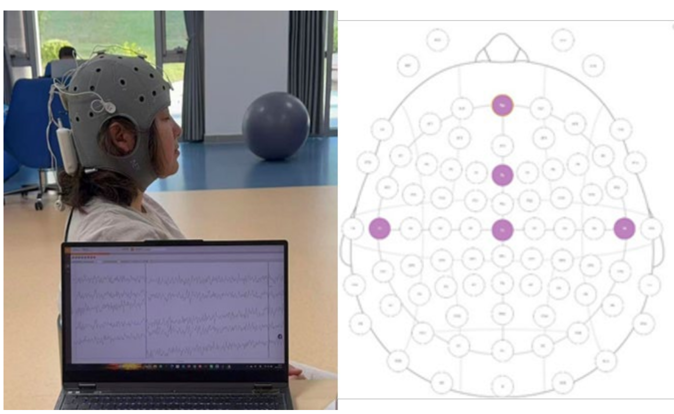

Figure 1. The subject is conducting an experiment (left) and location map of EEG data acquisition (right)

In the experiment, subject performed 50 catching-releasing tasks per group, with the tasks randomly ordered to reduce anticipatory EEG interference caused by fixed action sequences. A total of six groups of experiments were conducted for light and strong actions respectively. Based on the EEG equipment used, this study adopted five specific channels: Fpz, Fz, Cz, T7, and T8, positioned according to the international 10-20 system. After data cleaning, the final valid data consisted of 300 light-intensity action samples (Light Catching, LC: 150; Light Releasing, LR: 150) and 250 strong-intensity action samples (Strong Catching, SC: 125; Strong Releasing, SR: 125) for model training.

### B. EEG data preprocessing

A systematic preprocessing pipeline was implemented to extract multi-dimensional features from motor imagery signals while ensuring the replicability of the experimental protocol. To eliminate abnormal data, each 1.2-second EEG segment was split into 11 sub-segments (0.2s window, 0.1s step), and segments exceeding a manually set standard deviation threshold were excluded to avoid extreme noise interference.

The feature extraction follows a dual-pathway strategy to fully capture neural dynamics: the first pathway applies six groups of zero-phase Butterworth band-pass filters (0.5–4 Hz, 4–8 Hz, 8–12 Hz, 12–30 Hz, 30–45 Hz, and 0.5–45 Hz) to isolate fundamental rhythmic activities and Event-Related Desynchronization (ERD) patterns; simultaneously, the second pathway targets high-frequency oscillations by processing raw signals through four additional narrow-band filters (30–45 Hz, 60–80 Hz, 80–100 Hz, and 100–120 Hz) and employing the Hilbert transform to extract the instantaneous amplitude envelopes as supplementary energy-based features.

Consequently, by combining these 6 primary filtered bands with the 4 high-frequency envelopes, the preprocessed EEG data for each trial is represented as a 3D tensor of [5, 600, 10], where 5 denotes the physical electrode channels (Fpz, Fz, Cz, T7, T8), 600 denotes the time points, and 10 represents the unified frequency-based feature dimensions. Among them, the catching samples consist of LC and SC (150+125=275), and the releasing samples consist of LR and SR (150+125=275), with a total sample size of 550.

### C. Deep Learning Dataset Partition

To verify the universal improvement effect of the ‘Light-Strong’ Effort component in LBMS on motor imagery EEG decoding across different deep learning architectures, this study constructed Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) classification models respectively. Binary classification tasks were conducted using EEG data corresponding to subjects' ‘Light Catching (LC)’, ‘Light Releasing (LR)’, ‘Strong Catching (SC)’, and ‘Strong Releasing (SR)’ actions.

*a) Dataset Partitioning:* To establish a baseline for quantifying the feature gain of the "Light-Strong" dimension in the LBMS, a standardized data mixing process was first implemented to generate a baseline dataset without intensity distinction. During extraction, the middle segment of trials was selected to reduce systematic errors; in the mixing phase, light and strong data were first concatenated along the trial dimension, and then the trial order was shuffled using a random shuffle algorithm to eliminate sequence correlation in the intensity dimension.

The model evaluation followed a within-subject approach. To ensure scientific integrity and avoid data leakage, a trialwise split strategy was strictly adopted: the original trial-based data were first partitioned into a training set, a validation set, and a test set according to a ratio of 8:1:1 before any windowing augmentation was applied. Based on the total of 550 samples, this resulted in 440 samples for training, 55 for validation, and 55 for testing. This protocol ensures that all segments generated from a single trial remain within the same subset, preventing the model from overestimating performance by "seeing" overlapping temporal fragments during training. For signal refinement, the first 50 data points before action initiation were removed to eliminate trigger noise, retaining core action signals (start offset = 20, end offset = 0). Subsequently, a sliding window method segmented signals with a length of 400 points and a step of 100 points. Finally, Z-score normalization was performed across features to enhance training stability.

*b) LSTM Model:* The Long Short-Term Memory (LSTM) network was employed for feature extraction and classification of EEG signals, leveraging its inherent advantage in capturing temporal features. The optimized LSTM's input dimension was auto-adapted to preprocessed data features for full capture of EEG temporal dependencies. Cross-entropy loss was chosen for binary classification label distribution, while the momentum-augmented SGD optimizer (initial learning rate = 0.0008, momentum coefficient = 0.9) was used to speed up convergence. L2 regularization (weight decay = 0.003) was introduced to mitigate overfitting. A learning rate scheduler was applied to balance convergence speed and generalization: the rate was halved every 20 epochs during the 700-epoch training.

*c) CNN Model*: A Convolutional Neural Network (CNN) was constructed for comparative validation, with better performance in spatial feature extraction. Its input is a 3D EEG tensor (channels × time steps × frequency features), with a three-stage feature extraction structure: the first stage includes a large (3,5) convolutional layer (64 kernels, padding = (1,2)), batch normalization, ELU activation, frequency-specific (1,2) max-pooling, and Dropout (rate = 0.2) for long-term dependency capture and frequency-dimensionality reduction; the second stage uses a stacked dual (3,3) convolutional structure (128 kernels, padding = 1) with batch normalization and ELU activation per sublayer, followed by (2,2) max-pooling and Dropout (rate = 0.3) to enhance multi-scale feature fusion; the third stage comprises a (3,3) convolutional layer (256 kernels, padding = 1), batch normalization, ELU activation, (1,1) adaptive average pooling, and Dropout (rate = 0.5) to achieve deep feature refinement and dimensionality standardization. Pooling-compressed features output binary classification probability distributions via the Softmax layer. The CNN adopted the same training configuration as the LSTM.

*d) CNN-BiLSTM-Attention Model*: To achieve synchronous and accurate decoding of four types of motor intentions (LC, LR, SC, SR), a hybrid CNN-BiLSTM-Attention model was constructed by integrating spatial feature mining, temporal dependency capture, and key feature enhancement capabilities. The model adopts a three-stage feature processing architecture: the input is a 3D EEG tensor (number of channels × time steps × frequency features). Firstly, spatial features are extracted through two convolutional blocks — the first convolutional block includes a 3×3 convolutional layer (16 kernels, Same Padding), batch normalization, ReLU activation, and 2×2 max-pooling; the number of kernels in the second convolutional block is increased to 32 to extract high-level spatial features. After reshaping, the convolutional output is fed into a BiLSTM layer (hidden layer dimension = 128), where the bidirectional structure synchronously captures forward and reverse temporal dependencies, and inter-layer Dropout (dropout rate = 0.3) suppresses overfitting. Finally, a self-attention layer weights and highlights key temporal features, and the probability distribution of the four classes is output through a 64-dimensional fully connected layer and Softmax activation.

The training configuration is adapted to the four-classification task: the loss function uses Focal Loss with class weights (weights [1.3, 1.1, 1.5, 1.5] to match sample distribution, $\gamma = 2.0$); the optimizer is SGD (learning rate = 0.0006, momentum = 0.9, L2 regularization coefficient = 0.003); the batch size is 32, and the model is trained for 800 epochs. An early stopping strategy (training terminated if the validation set accuracy does not improve for 20 consecutive epochs) is adopted to ensure generalization ability.

## III. RESULTS

### A. LSTM Model Result

The LSTM model's test accuracy is presented in Figure 2. To establish a benchmark, mixed ‘Light-Strong’ action data were trained under the same protocol (also see Methods), yielding a baseline result of 0.87. For all ‘Catch-Release’ action combinations, the model attains an overall recognition accuracy of over 0.83, with cross-intensity combinations reaching 0.90 (SC/LR) and 0.95 (LC/SR). These results, being superior to the 0.87 baseline, verify the strong discriminability of EEG features induced by the ‘Light-Strong’ Effort dimension. This finding provides the LSTM model with a stable basis for temporal feature differentiation. Despite the complexity of multi-dimensional discrimination, cross-intensity combinations still achieve excellent performance, validating the value of the LBMS framework in enhancing motor imagery EEG decoding.

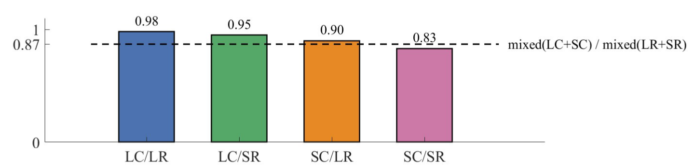

Figure 2. Test Accuracy of LSTM

### B. CNN Model Result

The CNN's test accuracy is shown in Figure 3. To provide a benchmark, mixed ‘Light-Strong’ action data were trained as a baseline, yielding a result of 0.88. The pattern confirmed the effectiveness of the ‘Light-Strong’ Effort component: for action combinations with different intensities, the model achieved recognition accuracies of 0.90 (LC/SR) and 0.95

(SC/LR). Notably, these cross-intensity results outperformed the 0.88 baseline and were superior to single-intensity combinations (LC/LR: 0.89; SC/SR: 0.87). This performance trend underscores that integrating the LBMS framework enhances the discriminability of EEG features. The CNN model's gains over the baseline and single-intensity tasks further validate the core value of the 'Light-Strong' Effort component in motor imagery EEG decoding.

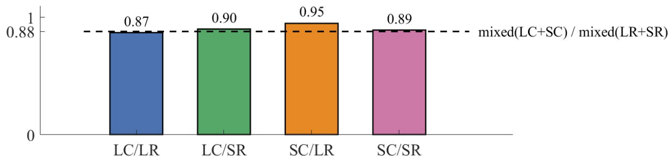

Figure 3. Test Accuracy of CNN

## C. *CNN-BiLSTM-Attention Model Result*

The classification results of the hybrid model are shown in Figure 4, with a maximum test accuracy of 0.96 and an average accuracy of 0.935. The integration of spatial and temporal features in the classification results addresses the one-sidedness of feature capture in single-architecture models. By weighting and focusing on EEG features during the key phases of movements, the attention mechanism effectively breaks through the bottleneck in distinguishing actions with a single intensity.

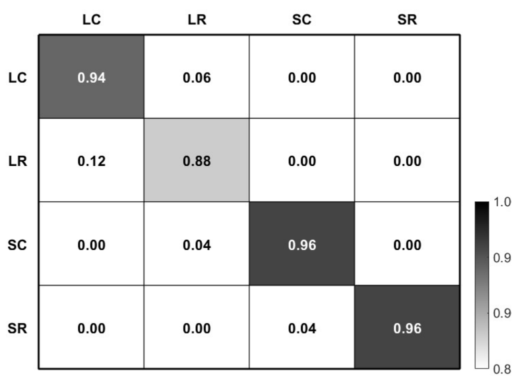

Figure 4. CNN-BiLSTM-Attention Normalized Confusion Matrix

# IV. CONCLUSION

The research results confirm a core premise: motor intentions with distinct 'Light-Strong' attributes in the 'Effort' component of the LBMS are distinguishable via EEG signal analysis. This verification responds to practical needs—human movements inherently involve force intensity variations in real scenarios, making the integration of this dimensional distinction into motor intention recognition practically necessary. Building on this confirmation, the proposed hybrid CNN-BiLSTM-Attention deep learning model achieves an average test accuracy of 0.935 and a peak accuracy of 0.96 in the four-class task (LC, LR, SC, SR), enabling effective discrimination of actions with different 'Effort' characteristics.

This achievement is expected to enhance the granularity of motor intention recognition, laying a theoretical foundation for BCI-driven devices to perform fine-grained operations.

Future work will focus on expanding the description of actions in other components of LBMS, exploring the improvement effect of multi-dimensional action semantics on the discriminability of EEG features. Additionally, the number of subjects will be increased to reduce sample randomness.

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